CSA0412-OPERATING SYSTEMS

EXP 09-Round robin scheduling

```
#include <stdio.h>

int main() {
    int n, i, tq, time = 0, remain;
    int bt[20], rt[20], wt[20], tat[20];
    float avgwt = 0, avgtat = 0;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    for (i = 0; i < n; i++) {
        printf("Enter burst time for P%d: ", i + 1);
        scanf("%d", &bt[i]);
        rt[i] = bt[i];
    }

    printf("Enter time quantum: ");
    scanf("%d", &tq);

    if (n <= 0 || n > 20 || tq <= 0) {
        printf("Invalid input\n");
        return 1;
    }

    remain = n;

    while (remain > 0) {
```

```
    for (i = 0; i < n; i++) {
        if (rt[i] > 0) {
            if (rt[i] > tq) {
                time += tq;
                rt[i] -= tq;
            } else {
                time += rt[i];
                wt[i] = time - bt[i];
                rt[i] = 0;
                tat[i] = time;
                remain--;
            }
        }
    }
}

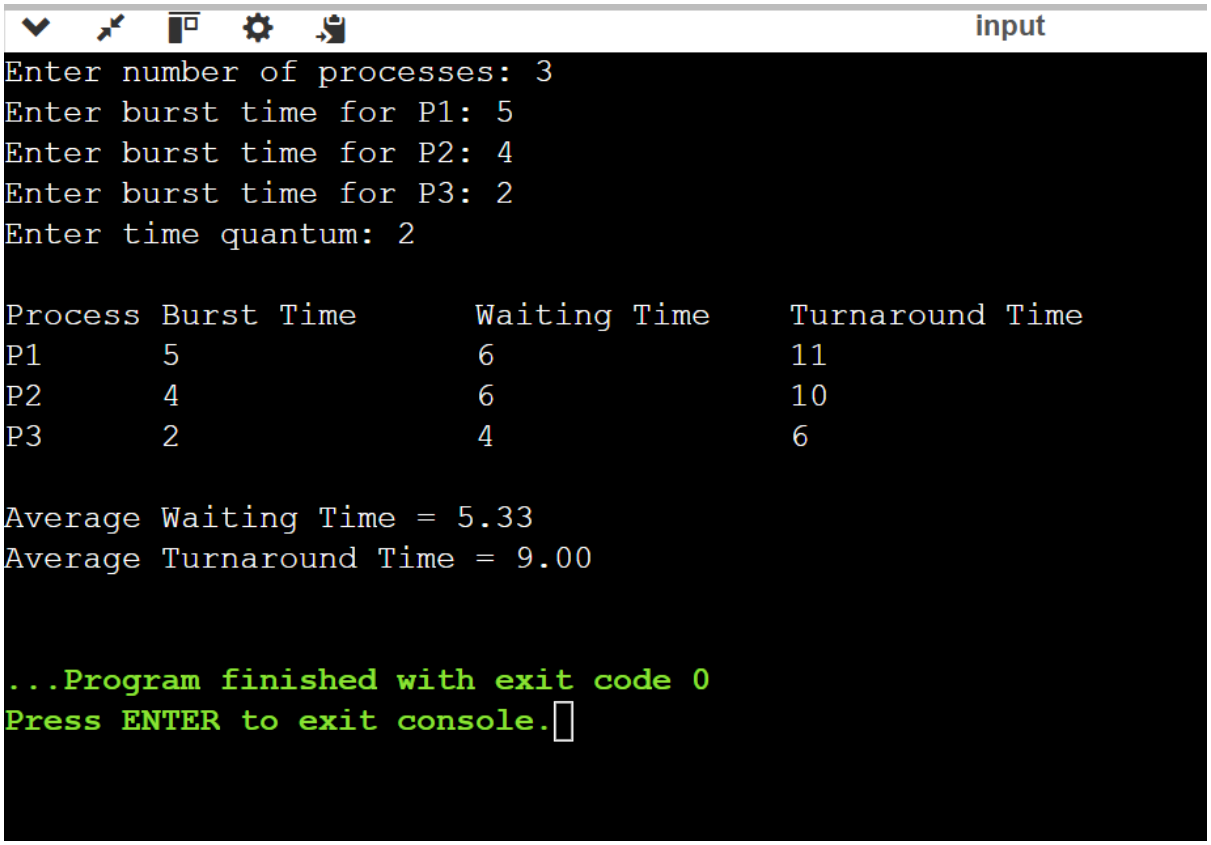
printf("\nProcess\tBurst Time\tWaiting Time\tTurnaround Time\n");

for (i = 0; i < n; i++) {
    printf("P%d\t%d\t\t%d\t\t%d\n",
           i + 1, bt[i], wt[i], tat[i]);
    avgwt += wt[i];
    avgtat += tat[i];
}

printf("\nAverage Waiting Time = %.2f\n", avgwt / n);
printf("Average Turnaround Time = %.2f\n", avgtat / n);

return 0;
```

}


input
Enter number of processes: 3
Enter burst time for P1: 5
Enter burst time for P2: 4
Enter burst time for P3: 2
Enter time quantum: 2
Process Burst Time Waiting Time Turnaround Time
P1 5 6 11
P2 4 6 10
P3 2 4 6
Average Waiting Time = 5.33
Average Turnaround Time = 9.00
...Program finished with exit code 0
Press ENTER to exit console.